Sagar Gupta

ProServe (Cloud Consultant) - DevOps/MLOps | Amazon Web Services (AWS) | Hyderabad, India

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Summary

ML Engineer with 2 years at AWS Professional Services as ProServe (Cloud Consultant) - DevOps/MLOps, building and shipping ML systems. Fixed a train/serve skew in a LeetCode rating model that had tripled its backtest error (MAE 38.76 to 12.46 on 57 contests). Architected medical image classification pipeline on SageMaker with blue-green endpoints, auto-scaling, and drift detection. Built a neural network framework from scratch in NumPy, backpropagation included. Upstream contributor to Feast feature store, Apache Airflow, and Prefect. 5x AWS certified including **AWS Machine Learning Engineer – Associate** and **AI Practitioner**. LeetCode Guardian, 2166 peak.

Experience

Amazon Web Services (AWS) Aug 2024 – Present
ProServe (Cloud Consultant) - DevOps/MLOps Hyderabad, India

- **MLOps Pipeline – SageMaker (SME program, Jun-Aug 2025):** Built the SageMaker pipeline that trains, gates and ships medical image classification models, with blue-green endpoint deployment, auto-scaling, and PSI drift detection that triggers retraining – zero-downtime updates with automatic rollback on error and latency alarms. Full IaC via Terraform + CodePipeline; API Gateway + Lambda inference API behind CloudFront frontend. Generalized and published by AWS as an official open-source sample (aws-samples, MIT-0).
- **Lead DevOps Consultant, Enterprise customer (Feb 2026 – Present):** Built a Terraform landing zone: Organizations OUs, Control Tower, SCPs, tag policies, account baselines and least-privilege deploy roles.
  - Automated account vending with Terraform Cloud and OIDC federation.
  - Automated network onboarding and tag-based firewall access with Transit Gateway.
- **DevOps Consultant, Global financial services customer (Oct 2024 – Aug 2025):** Wrote tested Terraform modules (terraform test, PyTest) for a multi-account landing zone. Worked on preventative security controls and EventBridge-based cross-account automation.
- **DevOps Consultant, Financial market infrastructure customer (Aug 2025 – Nov 2025):** Modernized legacy Terraform to cloud-native IaC patterns.
- **Internal Impact:** Published two more AWS samples, on Organizations governance and an AgentCore agent with governed MCP tools (see Publications). AWS Professional Services MLOps SME program graduate (2025). AWS technical field community ambassador (AI/ML, DevOps, CloudOps, Agentic AI, Serverless); Anthropic Claude Ambassador (2026). Associate Speaker Certification (Renewed 2026); Senior Speaker Certification (2026). Security content review qualification (2026). AWS Jam content tester, DevOps and GenAI (2026).

Amazon Web Services (AWS) Mar 2024 – Jul 2024
Associate Professional Services – DevOps (Internship) Hyderabad, India

- Migrated on-premises application to AWS ECS with blue-green deployment via Terraform, CodePipeline, and CodeDeploy – zero-downtime releases. Established observability with X-Ray and CloudWatch distributed tracing.

Ikarus-3D May 2023 – Jul 2023
Software Developer Associate (Internship) Mohali, India

- Containerized production application on AWS ECS/ECR with GitHub Actions CI/CD, and deployed SonarQube on EC2 + RDS to embed continuous code quality analysis into the delivery pipeline.

Publications & Recognition

- **Published by AWS:** aws-samples/sample-sagemaker-image-classification-mlops (author, MIT-0) – SageMaker MLOps in 86 Terraform files across 4 stacks and 13 modules: clinical quality and Fairlearn fairness gates, hourly PSI drift that starts retraining, canary deploys with auto-rollback. *2026*
- **Published by AWS:** aws-samples/sample-aws-terraform-org-governance (author, MIT-0) – Terraform for AWS Organizations governance on Control Tower: six policy types as 20 policy documents, delegated admin for 15 services. *2026*
- **Published by AWS:** aws-samples/sample-agentcore-harness-agent-terraform (author, MIT-0) – AWS architecture agent on the Bedrock AgentCore harness, all in Terraform (71 resources): 9 MCP tools behind a default-deny Cedar policy, served to Claude Code, Codex and Kiro; promoted to PROD only after an evaluation passes. *2026*
- **LeetCode Guardian** (top 1.1%) – 2166 peak rating, 1300+ problems, 107 contests. GeeksforGeeks 500+ problems; HackerRank 6-Star Problem Solving.

Industry Certifications (5)

AWS: Solutions Architect – Associate | Developer – Associate | Machine Learning Engineer – Associate
AWS: AI Practitioner | Cloud Practitioner

Personal projects and opinions are my own and do not represent Amazon or AWS.

Projects

Ledger Sync Jan 2026
React 19, TypeScript, FastAPI, SQLAlchemy, Neon PostgreSQL, Vercel GitHub | Live
◇ Self-hosted personal-finance app: imports .xlsx, .xls and .csv bank statements with SHA-256 transaction IDs so re-imports stay idempotent, then serves 26 analytics pages (Indian fiscal-year tax planning, investments, FIRE, multi-currency) and an AI assistant limited to 15 read-only tool calls. FastAPI on Vercel over Neon PostgreSQL, React on GitHub Pages, with 298 test files across both stacks.

orrery Oct 2026
Agent Skill, Claude Code Plugin, Node.js, ELK, SVG GitHub | Gallery
◇ Agent Skill that turns a YAML spec into animated architecture diagrams with 18,209 official icons (AWS, Azure, Google Cloud, Kubernetes, GitHub, IBM), each cloud pack rebuilt from the vendor's own download and pinned by checksum. ELK auto-layout, a validator whose every finding names the fix, and a review board of 3 reviewer agents; 40 reference diagrams are rendered and checked in CI. Runs in Claude Code, Codex, Cursor, Gemini CLI and Kiro.

LeetCode Rating Predictor Dec 2023
Python, TensorFlow, NumPy, FastAPI, React, Docker GitHub | Live
◇ Rating model on 15 engineered features. Found and fixed a train/serve skew on participant count that had tripled backtest error (MAE 38.76 to 12.46 on 57 contests). Moved serving to NumPy-only inference pinned by a golden test, cutting the runtime from ~1.8 GB to ~65 MB; a weekly job refreshes the data and re-runs the backtest.

Bedrock Multi-Model MCP Apr 2026
TypeScript, MCP Protocol, AWS Bedrock Converse API GitHub
◇ MCP server putting 26 text models from 7 providers, plus image, video and embedding models, behind 8 tools and one set of AWS credentials. Benchmarked in us-east-1: 26 of 26 models answered and 156 of 156 calls succeeded.

itr-agent Jul 2026
TypeScript, MCP Protocol, Node.js, Zod GitHub
◇ Local-first MCP server for Indian income tax: the LLM interviews and calls 12 tools, while a deterministic engine over a versioned JSON rule pack computes every rupee. 151 tests pin it to published worked examples and the statute, on an Ubuntu and Windows CI matrix.

Other Notable Projects: Claude Cost Optimizer (cache-aware Claude cost model), GroundZeroNN (neural network from scratch in NumPy), GitScope (Chrome Web Store extension).

Open Source Contributions (11 merged upstream)

- **Apache Airflow** (47K☆) – template_fields on SalesforceBulkOperator, so its arguments render via Jinja at run time (#63109)
- **axios** (109K☆) – AxiosError message now built from AggregateError sub-errors, ending blank dual-stack errors (#11059)
- **lucide** (25K☆) – Fixed a TypeError crash when Vue templates pass the kebab-case icon-node prop (#4512)
- **multer** (12K☆) – Files rejected by fileFilter no longer use up maxCount and fail later valid uploads (#1426)
- **terraform-aws-modules/atlantis** – Fixed custom ECS security group egress rules being silently replaced by allow-all (#446)
- **Also merged:** PrefectHQ/prefect, feast-dev/feast, awslabs/agent-plugins (#132, #212), awslabs/mcp, cloudposse/terraform-aws-tfstate-backend; co-author of the aws-ia/terraform-aws-vpc v4.8.0 fix.

Education

National Institute of Technology Warangal, Telangana Sep 2021 – Jul 2024
Master of Computer Applications (MCA) | NIMCET AIR 208 CGPA: 8.38

Sanskar College of Professional Studies (DAVV University), Indore May 2018 – Aug 2021
Bachelor of Computer Applications (BCA) CGPA: 7.34

Technical Skills

ML & Deep Learning: TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas; Dense Networks, CNNs, LSTMs, Ensemble Methods, Feature Engineering, Model Evaluation (MAE/RMSE, precision/recall)
MLOps on AWS: SageMaker (Training, Pipelines, Endpoints, Custom PSI drift jobs), Blue-Green Deployments, Model Versioning, CodePipeline, Feast Feature Store
GenAI & LLMs: Amazon Bedrock, Amazon Bedrock AgentCore, RAG Pipelines, LLM APIs (Anthropic, OpenAI), MCP Protocol, Prompt Engineering, Claude Code, Amazon Q, Kiro
Serving & Infrastructure: FastAPI, Lambda, API Gateway, CloudFront, Network Firewall, AWS Backup, Systems Manager Patch Manager, Docker, Terraform, GitHub Actions, GraphQL
Languages: Python, SQL, TypeScript, JavaScript, HCL, C++, Java

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